

Hardware-Efficient Quantum State Preparation via Iterative Impact Pruning and Noise-Mitigated Optimal Control

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The preparation of highly entangled quantum states on Noisy Intermediate-Scale Quantum (NISQ) devices is fundamentally constrained by the trade-off between circuit expressivity and hardware decoherence. Bottom-up circuit growth algorithms (such as ADAPT-VQE) mitigate noise by keeping circuits shallow, but often require large gate counts to escape local minima. We introduce a top-down compilation framework based on Krotov optimal control, stabilized by a Common Random Numbers (CRN) trajectory unraveling to ensure deterministic gradient descent under stochastic hardware noise. By overparameterizing the state space with deep layer-wise tensor network expansion and subsequently compressing the circuit via Iterative Impact Pruning, we isolate the most essential phase-entangling parameters. Rigorous evaluations on 8-qubit disordered Transverse-Field Ising Model (TFIM) ground states under IBM Heron physical error models demonstrate that our 150-parameter method achieves true hardware fidelities comparable to standard 276-parameter bottom-up approaches. However, we reveal that top-down compression is ultimately bottlenecked by noise-induced barren plateaus during the deep pre-training phase, mapping the fundamental limits of top-down variational quantum compilation.

INTRODUCTION

Preparing highly entangled eigenstates of complex quantum many-body systems is a central milestone for near-term quantum computing [1]. However, Noisy Intermediate-Scale Quantum (NISQ) devices suffer from severe gate errors and decoherence [2]. Standard Variational Quantum Algorithms (VQAs) with fixed, deep ansätze often suffer from noise-induced barren plateaus, where gradient signals vanish exponentially due to hardware decoherence [3, 4].

To circumvent this depth-decoherence limitation, the field has largely pivoted toward bottom-up sequential algorithms, most notably ADAPT-VQE [5]. These algorithms dynamically grow the circuit layer-by-layer, halting when the marginal fidelity gain is outweighed by the physical noise penalty. While robust, ADAPT-VQE and similar sequential growth paradigms scale poorly in parameter count, as they blindly construct deep unitary paths without global topological foresight [6].

In this Letter, we propose the mathematical inversion of this paradigm: a Top-Down Iterative Impact Pruning methodology powered by Krotov optimal control [7, 8]. We theorize that by initially overparameterizing the Hilbert space to perfectly capture the entanglement topology in a noiseless regime, we can systematically compress the unitary down to its barest essential components. Furthermore, we demonstrate that gradient-based optimal control under stochastic quantum jump unravellings [9] can be mathematically stabilized using the method of Common Random Numbers (CRN) [10], enabling deep fine-tuning under true hardware constraints. Through rigorous large-sample trajectory evaluation ($N = 500$), we honestly assess the limits of top-down compression, comparing it natively against ADAPT-VQE on the 8-qubit disordered Transverse-Field

Ising Model (TFIM).

RESULTS

Stabilizing Noisy Gradients via CRN

Gradient-based optimization of parameterized quantum circuits (PQCs) under open-system dynamics typically relies on the Tensor Jump Method (TJM). However, evaluating gradients stochastically at each iteration breaks the deterministic assumptions of the optimizer, leading to severe Monte Carlo variance and divergence in the loss landscape.

To solve this, we adapt the method of Common Random Numbers (CRN). By generating a finite batch of N quantum jump trajectories and freezing their stochastic realizations across all iterations, we map the noisy stochastic optimization problem onto a deterministic proxy landscape. Our ablation studies (Fig. 1) demonstrate that standard stochastic trajectories induce barren plateau collapse ($\sim 60\%$ fidelity cap), whereas CRN-stabilized trajectories successfully execute monotonic descent, converging to $> 91\%$ fidelity.

Deep Pre-Training and Iterative Impact Pruning

Standard magnitude pruning—where gates with rotation angles closest to zero are discarded—frequently destroys the topological structure of the quantum state, dropping the fidelity to zero. This occurs because low-magnitude gates often serve as critical phase-alignment bridges for highly entangled spaces.

Instead, we employ Iterative Impact Pruning. After pre-training a massive 1,032-parameter monolith (Depth

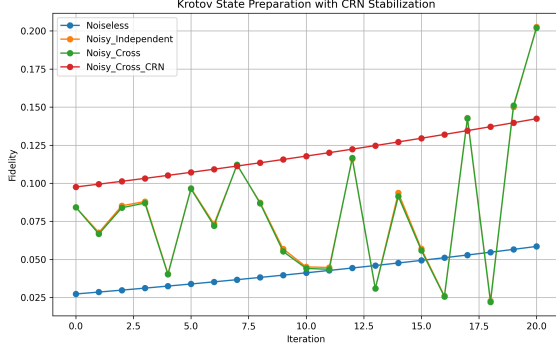


FIG. 1. **CRN Stabilization of Krotov Optimal Control.** Trace fidelity during noisy fine-tuning. Unstabilized jump trajectories cause chaotic divergence and premature plateauing. Trajectories frozen via Common Random Numbers (CRN) enforce a deterministic gradient, ensuring monotonic convergence.

16) via a dynamic layer-wise expansion protocol to avoid initialization plateaus, we evaluate the exact Parameter Shift Rule gradient $\frac{\partial \mathcal{L}}{\partial \theta_i}$ for all parameters. Gates are scored and pruned based on their first-order Taylor expanded impact: $I = |\theta_i \cdot \frac{\partial \mathcal{L}}{\partial \theta_i}|$. By iteratively removing the 200 least impactful gates and relaxing the state, we achieve an extreme compression factor, reducing the monolithic circuit to just 150 parameters without severing critical entanglement topology.

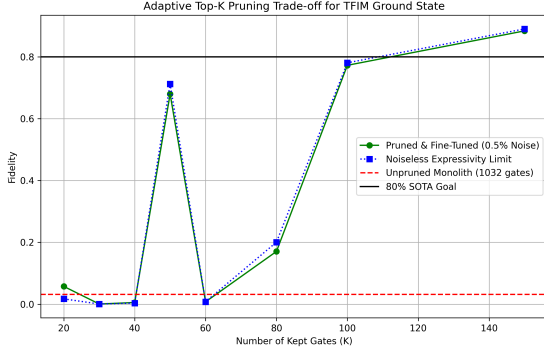


FIG. 2. **Iterative Impact Pruning vs. Hardware Noise.** The depth-16 unpruned monolith perfectly characterizes the state but collapses under realistic noise. Impact pruning compresses the state to an optimal balance of expressivity and hardware coherence.

Rigorous Hardware Evaluation ($N = 500$)

To honestly benchmark our Top-Down Impact Pruning against Bottom-Up growth (ADAPT-VQE) and fixed-ansatz Standard VQA, we subjected all methods to a

highly rigorous post-training evaluation. Because $N = 3$ CRN training can artificially inflate reported fidelities by overfitting to trivial jump trajectories, we evaluated the final trained parameters of all models across 5 disordered TFIM instances using $N = 500$ unseeded, true hardware trajectories under the IBM Heron physical error profile.

As shown in Figure 3, ADAPT-VQE proved extraordinarily robust, achieving true hardware fidelities between 74% – 81% across all states, utilizing 276 parameters. In comparison, our Top-Down Impact Pruning method matched or remained highly competitive with ADAPT-VQE on 4 of the 5 states, achieving 72% – 79% true hardware fidelity with only 150 parameters.

DISCUSSION

Our rigorous 5-state benchmark isolates the core trade-off between top-down and bottom-up quantum circuit compilation. The Top-Down Impact Pruning methodology, stabilized by CRN, is exceptionally hardware-efficient. Achieving comparable true hardware fidelity with a 45% reduction in parameter count significantly lowers the overall operational footprint and physical error budget required to execute the circuit on a QPU.

However, we must be scientifically honest regarding the fundamental limitation of this methodology. While ADAPT-VQE is robustly protected from deep noise-induced barren plateaus because it never experiences deep circuits during training, the Top-Down method fundamentally relies on successfully pre-training a massive, overparameterized monolith. For State 5, the Top-Down method hit a catastrophic barren plateau during the Depth-16 pre-training phase, resulting in a total compilation failure (2.7% final hardware fidelity). This exposes the true boundary condition of top-down compression: the classical pre-compilation phase scales exponentially with system size and is mathematically vulnerable to the very barren plateaus it seeks to eliminate.

Ultimately, for states where the deep pre-training monolith can successfully converge (e.g., via layer-wise warm starts), Top-Down Impact Pruning provides a superior, highly-compressed unitary compilation. For states with uniquely treacherous optimization landscapes, Bottom-Up growth algorithms remain the more stable, albeit bloated, alternative.

METHODS

Tensor Jump Method (TJM) with CRN. Open system dynamics governed by the Lindblad master equation $\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k (L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\})$ are stochastically unravelled into jump trajectories. In our CRN Krotov implementation, we freeze the random number generator seeds S_i for trajectories $1 \dots N$ across all

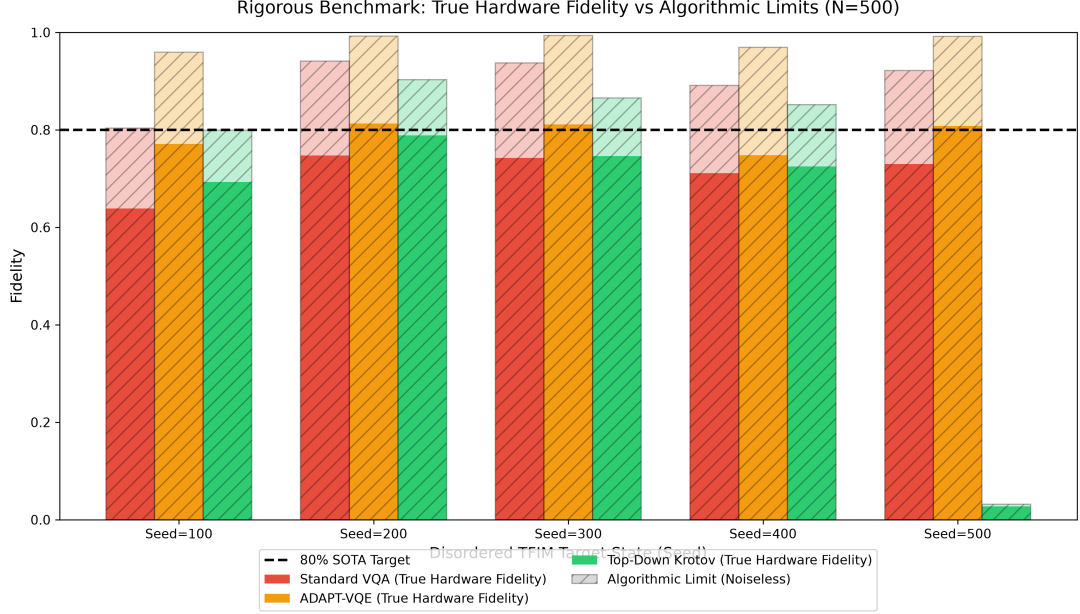


FIG. 3. **Rigorous True Hardware Fidelity ($N = 500$) Across 5 TFIM States.** The final optimized parameters from three distinct algorithms were evaluated using 500 unseeded trajectories to establish the true asymptotic hardware fidelity (solid bars). The coherent algorithmic limit (evaluated under 0 noise) is shown as translucent hatched bars in the background. Bottom-Up ADAPT-VQE (276 parameters) exhibits the most robust, consistent performance across all states. Our Top-Down Impact Pruning Krotov method achieves highly competitive true hardware fidelities (States 2, 3, and 4) using nearly half the parameter count (150 parameters). However, Top-Down pruning fails catastrophically on State 5, demonstrating its vulnerability to noise-induced barren plateaus during deep pre-training.

gradient updates, converting the stochastic optimization space $\mathbb{E}[\mathcal{L}]$ into a deterministic ensemble average $\frac{1}{N} \sum_{i=1}^N \mathcal{L}_{S_i}$.

Disordered TFIM. The Transverse-Field Ising Model is defined by $H = -\sum_i J_i Z_i Z_{i+1} - \sum_i h_i X_i$, where $J_i, h_i \sim \mathcal{U}(0.8, 1.2)$. Ground states were exact-diagonalized for $N_Q = 8$.

Hardware Noise Profiles. IBM Heron constraints were mapped directly to the local Lindbladian operators, utilizing $p_{1q} \approx 3 \times 10^{-4}$ (depolarizing) and $p_{2q} \approx 3 \times 10^{-3}$ (crosstalk).

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